

Suraj Chauhan

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Education

BSMS, Indian Institute of Science Education and Research, Bhopal	2021 Dec – 2026 Apr
Secondary High School, AMU City School 93.4	2020 Mar Aligarh, India
High School, SVM Senior Secondary School 95.2	2018 Aligarh, India

Research Experience

Comparative Gut Microbiome Analysis in Primates Lineage, <i>Prof. Vineet Kumar Sharma</i> ✉	2025 May – Present Meta Bio-Sys lab
• Large-scale metagenomic analyses of 5200+ WGS samples of NHP and humans to investigate the gut microbiome. • Metagenomic Reconstruction & Genome-Resolved Analysis: Performed metagenomic assembly and binning to reconstruct newly identified 2500+ metagenome-assembled genomes (MAGs), enabling investigation of host-microbiome interactions and previously unresolved microbial diversity. • functional annotation to identify lineage-specific Biomarkers using including antimicrobial resistance, vitamin biosynthesis pathways, bile acid metabolism, and CAZymes, distinguishing evolutionary adaptations from disease-associated dysbiosis.	
Human perception on free ranging dogs in India, Dr Anindita Bhadra	2024 May – 2024 Aug Dog Lab, IISER Kolkata
• The project examines human perceptions of free-ranging stray dogs (FRDs) across India through experimental surveys and data analysis. • Aims to understand public attitudes and behaviors towards stray dogs by gathering comprehensive insights from different regions of India. • The study has gained national and international recognition for its innovative approach to studying human-dog interactions.	
DNA Nanotechnology based RAM, Prof Kuntal Roy	2023 May – 2023 Jul Dr. Roy's Lab
• The project focused on developing a DNA-based storage device using nanopores and by combining molecular science and nanoelectronics to create next-gen RAM. • Hands-on experience was gained with key tools like AFM, XRD, Nanodrop, Spin Coater, Annealer, BCP Lithography and NMR. • All these techniques are performed to understand the topography and structure of DNA. Results were analyzed to generate observations and hypotheses for more feasible methods for future research.	
Biocuration of Rare-disease, Prof Parthiban Srinivasan	2022 Dec – 2023 Jan In -machino
• Curated genetic and clinical datasets on rare diseases including Hashimoto encephalopathy, analyzing gene-protein interactions, molecular pathways, and potential therapeutic targets using BLAST and AlphaFold for comparative genomics and protein structure-function prediction. • Proposed gene-editing-based intervention strategies through computational analyses to support precision medicine and drug discovery efforts. • Acted as a bridge between the biological curation team and the technical development team, translating biological insights into structured inputs for algorithm development.	

Academic Projects

Investigating the impact of cell lines on HIV-1, Prof. Nagarjun Vijay The impact of altered CD3G expression in T lymphocytes on immune function and its potential implications in the context of viral infections, specifically HIV-1 using single-cell RNA-seq using Seurat
CRISPR-Evasive Lytic LE1 Phage for Targeted Leptospira Elimination, Prof. Vikas Jain Modifying LE1 from a lysogenic to a lytic phage and incorporating anti-CRISPR genes (AcrIIA4 and AcrIF7) will inhibit Leptospira's CRISPR-Cas systems and defence islands, increasing the phage's ability to infect and lyse the bacteria effectively.
Investigating sneezing pattern in free ranging dogs, Prof. Vinita Gowda Conducted a behavioral study on free-ranging urban dogs to investigate the context and potential communicative role of sneezing. Used structured observations and ethogram-based coding to analyze sneezing frequency across social and solitary settings. Preliminary findings suggest sneezing may aid in social coordination or communication

How Social Competition Affect True and False Recognition ?, Prof. Aditi jublie

This study explores how competitive group settings influence true and false memory recognition. Results reveal that competition reduces overall accuracy but also lowers false memories by enhancing detail-focused recall.

Skills

Lab

PCR, Gel electrophoresis, DNA/RNA extraction
Cloning, Western Blotting and Sanger sequencing

Microscopy and other Techniques

AFM, XRD, TEM, SEM, Confocal
Nanodrop, NMR and Fluorescence

Coding and Data Analytics tools

Bash, UNIX C, Python, Perl, PyMol, and R

Video Editing

Da Vinci, Capcut, Premier pro

Visualization tools

Bio-render, Chemdraw, Matlab

Design

Canva, MS Power Point, Gamma

MS office

Power point, Word, Excel

Data-Analysis

Tableau, Power BI

Personal Projects

Whole-Exome Sequencing (WES) Analysis

A reproducible, end-to-end pipeline for Whole-Exome Sequencing analysis, from raw SRA reads through aligned BAM, base recalibration, variant calling, filtering, and functional annotation.

Spatial Transcriptomics Analysis

A comprehensive, reproducible workflow for spatial RNA-seq data analysis using Python-based tools (Squidpy, Scanpy, Spatialdata)

ATAC-Sequencing Analysis

This repository contains a reproducible workflow for analyzing 10x Genomics single-cell ATAC-seq (scATAC-seq) data using the Signac + Seurat framework in R.

Single-Cell RNA-seq Analysis with Scanpy

A complete pipeline for single-cell RNA sequencing (scRNA-seq) data analysis using [Scanpy](#) , an open-source Python toolkit for analyzing single-cell gene expression data.

Bulk-RNA-Sequencing Analysis

A comprehensive workflow for analyzing bulk RNA-sequencing data from raw FASTQ files to differential gene expression and functional enrichment analysis.

Libraries and tools

Meta-Genomics

SRA toolkit, FastQC, SeqKit, Trimmomatic, Bowtie2, Kraken2, Bracken, MetaPhlAn, Megahit, Prodigal, CdHit, dbCan, MetaWrap, Metaquast, Drep, Coverm, CheckM, and Salmon

Meta-transcriptomics

fastp, SortMeRNA, Barrnap, RiboPicker, BBDuk, HISAT2, Kaiju, HUMAnN3, DIAMOND, eggNOG-mapper, Salmon, Kallisto, DESeq2, edgeR, and ALDEx2.

RNA-Seq

Scanpy, Seurat, STAR, HISAT2, DESeq2, edgeR, limma, clusterProfiler

Leadership and Community Services

Member, Animal Welfare Committee

2025 Jun – Present

Established a committee and designated feeding zones for canines
Collared and vaccinated 40+ dogs within the campus premises

Secretary, Science Council

2023 Oct – 2024 Aug

- Devote 250+ hours throughout the entire academic year
- Managed all the STEM clubs, social initiatives, events, seminars and talks
- Established science media centre as an Outreach and Communication Body

Convenor, Singularity 23-24

2023 Dec – 2024 Feb

- Devotes 200+ Hours for Planning and Execution of 7 days Conference
- Organized five talks, 7 Workshops, 20 Events and Social Initiatives in Schools
- Led team of 90 members, i.e Event, Logistics, Outreach and Workshop teams

Senate Member, Student Activity Council

2023 Apr – 2024 Aug

- Student representation, decision-making, Policy making and Advocacy

- Represent the student body in the the Academic Senate meeting.
- Led meeting to plan community service initiatives

Awards and Exams

CSIR UGC-NET, NTA Qualified the test	2025 Aug 23
GATE EY, All India Rank : 176 Nationwide test for admission for Masters and Doctorate programmes	2025 Mar
Silver Medal, Inter college sports meet, Kho-Kho	2023 Dec 26
KVPY SB, All India Rank : 106 Prestigious scholarship program to pursue research careers for undergraduate studies	2022 Jan 22
Gold medal, Silver Zone Science Olympiad	2021 Mar 10
Best Achiever, AMU City School Given for exceptional performance in academia and extra curricular activity.	2020 Mar 10
Gold Medal, National Science Congress For the project on environment and sustainability in nationwide program	2018 Feb 12

Languages

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| • English | • Hindi |
| • Urdu | • Sanskrit |
| • German | |

Relevant Coursework

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| • Learning Memory and Cognition | • Evolutionary Ecology |
| • Advances in Omics and Microbiology | • Biochemistry and Biomolecules |
| • Recombinant DNA Technology | • Bioinformatics |
| • Biophysics and Bioinstrumentation | |

Reference

Prof. Vineet K Sharma, Principal Investigator	vineetks@iiserb.ac.in IISER, Bhopal
Prof. Debasis Nayak, faculty Advisor	debasis@iiserb.ac.in IISER, Bhopal
Prof. Aasheesh Srivastava, faculty Advisor	asrivastava@iiserb.ac.in IISER, Bhopal